# Design and analysis of experiments

HU | Spring 2026 (Winter term 25-26)

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**Abstract**

The course addresses the design and analysis of experiments. It is a hands-on course in which theoretical results are introduced through lecture, demonstrated in practice, and worked on in groups. Topics include randomization schemes, analysis strategies, design evaluation, and workflows.

## 1 Class resources

- All resources: https://macartan.github.io/ci
- This syllabus: https://macartan.github.io/ci/experiments_syllabus_2026.pdf
- Slides: https://macartan.github.io/ci/experiments_2026.html
- Student survey
- Git repo https://github.com/macartan/ci: This repo is for both my causal inference class and this experiments class.

## 2 Times and locations

Meetings will be held at the WZB

| Date | Time | Location | Topic |
|---|---|---|---|
| 9 Jan 2026 | 9 - 13 | WZB A 305 | Getting started |
| 16 Jan 2026 | 9 - 13 | WZB A 305 | Doing experiments |
| 30 Jan 2026 | 9 - 13 | WZB A 305 | Causality and causal inquiries |
| 6 Feb 2026 | 9 - 13 | WZB A 305 | Answer strategies |
| 20 Feb 2026 | 9 - 13 | WZB A 305 | Data strategies and design evaluation |
| 27 Feb 2026 | 9 - 13 | WZB B 001 | Topics: downstream experimentation, spillover experiments, patient preference trials, survey experiments |

The sessions will be structured roughly as follows.

- 9:00 - 11:00: lecture or discussions of readings
- 11:00 - 11:15: break
- 11:15 - 13:00: discussions, team presentations and in-class exercises

## 3 Expectations

### 3.1 Required

- Do the readings and take part in discussion (30%)
- Take part in field experiment I (30%)
- Design an experiment as a candidate for survey experiment II (40%)

## 3.2 Optional

- Prepare a research design or short paper, perhaps building on existing work. Typically this contains:
  - a problem statement
  - a description of a method to address the problem
  - analytic or simulation based results describing properties of the solution
  - a discussion of implications for practice.
  - A passing paper will illustrate subtle features of a method; a good paper will identify unknown properties of a method; an excellent paper will develop a new method.

# 4 Sessions and readings

The course combines lectures, discussion, and activities.

All readings are linked from this document.

I will draw material especially from Blair, Coppock, and Humphreys (2023): Research Design in the Social Sciences

Recommended but not open access: Gerber and Green (2012) with supplementary material available here: https://isps.yale.edu/FEDAI

Readings for each session are given below.

## 4.1 Intro

Aims:

- Introduce the class
- Practice running experiments by hand
- Introduce and discuss field experiment I
- To prepare for next session: proposed adaptation of field experiment 1

**Course outline, tools**

## 4.2 Experimentation

Aims:

- Review lifecycle of an experiment (discussion)
- Discuss ethical challenges to experimentation
- Discuss a set of strong experiments (all graduate student projects)
- Develop group field experiment 1

**To do before class**: 1. the readings; 2. develop specific proposals in small groups for field experiment 1, completing individually or collectively a copy of the EGAP research design form

Required reading:

- The Hijab penalty and the Appendix
- Johnson: Protecting the Community: Lessons from the Montana Flyer Project
- Design lifecycle Blair, Coppock, and Humphreys (2023) p 319 - 358
- Example 1: Gerber, Green, and Larimer (2008)
- Example 2: Lowe (2021)

Optional reading:

- Humphreys (2015) on ethics in experiments
- DD, Ch 2 introduces ideas at a high level
- Alvarez, Key, and Núñez (2018)
- Humphreys, De la Sierra, and Van der Windt (2013) on fishing and registration
- Kurzban, Tooby, and Cosmides (2001)
- Mousa (2020)

## 4.3 Causality

Aims:

- Understand the potential outcomes framework
- Understand identification challenges and the role of randomization
- Understand varieties of estimands

**To do before class**: 1. the readings; 2. collectively complete a pre-analysis plan and ethics application for Experiment 1

Required:

- Holland is a beautiful classic reading on the fundamental problem of causal inference
- DD, Ch 7

Optional:

- II Ch 2, goes over both potential outcomes and DAGs
- Keele (2015) gives a good high level overview of many key ideas in this course
- Imai, King, and Stuart (2008)
- Hernán and Robins (2024) (Section 3.1, 7.2, 8.4, 10.1)
- Hernán and Robins (2024) Ch 6
- II Ch 4
- Lundberg, Johnson, and Stewart (2021)

## 4.4 Answer strategies

Aims:

- Understand alternative strategies to estimating treatment effects
- Understand design based strategies for estimating standard errors and $p$-values
- Understand confidence intervals
- Understand the role of covariates in experimental design

**To do before class**: 1. Collective presentation on the implementation and findings from Experiment 1

Optional:

- Freedman (2008) helps make connections between design based inference and regression
- Bind and Rubin (2020) on RI
- Lin (2012) relieves some worries you might have after reading Freedman (2008)
- Chang, Middleton, and Aronow (2024) follows up on Lin (2012).

## 4.5 Experimental Design and Evaluation

Aims:

- Understand alternative assignment strategies
- Understand statistical power
- Understand how to assess alternative diagnosands

**To do before class**: 1. The readings 2. Share initial design for for survey based experiment.

Required:

- DD, Ch 7
- DD, Ch 8 on data strategies
- DD, Ch 9 on answer strategies
- DD, Ch 13 gives more practical guidance on getting started

## 4.6 Topics

Aims: * Topics TBD

**To do before class**: 1. The readings 2. Round-robin report on design performance. Teams receive a class-mate's design, declare a model that produces possible data from the design (without consulting the authors design declaration, if any) and applies the original authors answer srtategy, reporting performance.

Required reading (TBD):

#### 4.6.1 downstream experimentation

- Coppock and Green (2016)

#### 4.6.2 spillover experiments

- Sinclair, McConnell, and Green (2012)
- Aronow and Samii (2017) on spillovers

#### 4.6.3 patient preference trials

- Benedictis-Kessner et al. (2019)

#### 4.6.4 survey experiments

- Hainmueller, Hopkins, and Yamamoto (2014)

#### 4.6.5 mediation experiments

- Imai, Keele, and Tingley (2010) on mediation

# 5 Prerequisites and tools

You should already have background in statistics up to the point of feeling comfortable with regression.

In addition you should know some `R`. Really, the more you can invest on getting on top of `R` before the class the better.

Before the first class please make sure your `R` is up-to-date and that you are working in `R studio`. Then make sure you have the following packages installed.

```
pacman::p_load(
  rstan,
  dagitty,
  DeclareDesign,
  CausalQueries,
  tidyverse,
  knitr
  )
```

Resources:

- Resources for learning `R`: http://www.r-bloggers.com/how-to-learn-r-2/
- Hadley Wickham's R for Data Science
- Grant McDermott on data science

## 5.1 Writing with `qmd`

Please plan to do drafting in `Quarto`. This is a simple markup language that lets you integrate writing and coding. This document is written in `Quarto` and the slides will be also.

The key thing is that you can insert code chunks like this.

```
# Define a random number
x <- rnorm(1)
```

Code like this is run as the document compiles, and results can be accessed as needed, like this: we just sampled the random number $x = 0.5712688$.

- See Wickham, Çetinkaya-Rundel, and Grolemund (2023) on Workflow and Quarto

I recommend using `Rstudio` as an editor. More information here: https://quarto.org/docs/get-started/hello/rstudio.html

## 5.2 AI

AI can be a great aid to help understand this material and also to develop code. However it should only be used as an aid: it makes many many mistakes; you remain responsible for your code and more importantly for your comprehension.

# References


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